

Emily Foley
PhD Student, University of Arizona
emfoley@arizona.edu

EDUCATION

University of Arizona

Doctor of Philosophy, Applied Mathematics

Tucson, AZ

Aug 2024 – Present

Wake Forest University

Bachelor of Science, Physics and Mathematics

Magna cum laude, Phi Beta Kappa

Winston-Salem, NC

Aug 2020 – Jun 2024

RESEARCH EXPERIENCE

Graduate Research

Project: Protostellar Outflows in Binary and Multiple Systems

Advisor: Kaitlin Kratter

University of Arizona

2025 – Present

- Analyze protostellar jets in binary and multiple systems using Lagrangian numerical magnetohydrodynamic simulations
- Identify bound multiple systems and characterize outflow morphology and alignment relative to stellar spins and magnetic fields
- Develop analysis pipelines to generate synthetic velocity maps and assess how projection effects complicate observational interpretation of outflows

Undergraduate Thesis, Physics

Project: Investigating the Unruh State in 4D Schwarzschild-de Sitter Space

Advisor: Paul Anderson

Wake Forest University

2022 - 2024

- Analytically and numerically investigated quantum effects of black holes related to the Hawking Effect, focusing on Boulware and Unruh vacuum states in the cosmological region
- Numerically computed scattering states of Unruh radiation from black holes and compared their behavior to radiation emitted from the cosmological horizon

Undergraduate Thesis, Mathematics

Project: Spatiotemporal Chaos in the Damped-Driven Sine Gordon Equation

Advisor: John Gemmer

Wake Forest University

Sept 2023- Apr 2024

- Numerically characterized solutions to the damped and forced sine-Gordon equation, searching for spatiotemporal chaos
- Determined regions in parameter space in which solutions exhibit aperiodic late time behavior using numerical simulations in MATLAB

DOE Summer Undergraduate Laboratory Internship (SULI)

Project: Space Charge Calibration the ProtoDUNE Cryostat using a UV Laser System

Advisor: Sowjanya Gollapinni

Los Alamos National Lab

May 2023 - Jul 2023

- Developed an extensive analysis framework in python to correct electric field distortions in the detector for a prototype for the Deep Underground Neutrino Experiment (DUNE)
- Generated electric field correction maps using delaunay triangulation to map over 1 million Monte Carlo-simulated space charge distortions to simulated laser tracks

Research Experience for Undergraduates (NSF-REU)

Project: Updated Multimessenger Implications for Forthcoming LIGO Observing Runs

Advisor: Michael Coughlin

University of Minnesota

Jun 2022 – Aug 2022

- Conducted approximately 10,000 Monte Carlo simulations of gravitational wave signals to predict detection rates and sensitivity constraints for the next observation run of the LIGO detector network
- Simulated 3000 light curves using Monte Carlo radiative transfer code
- Results were used to inform data-driven proposals by the scientific community to detect electromagnetic counterparts to gravitational wave signals

FUNDING, AWARDS, AND HONORS

DOE Computational Science Graduate Fellowship University of Arizona	2025–Present
University Fellows Award University of Arizona	2024–Present
William E. Speas Award Wake Forest University	2024
APS Outstanding Undergraduate Oral Presentation American Physical Society	2023
Barry Goldwater Scholarship Nominee Wake Forest University	2023
Stamps Scholarship Wake Forest University	2020–24
Joint Math Meetings Travel Funding American Mathematical Society	2024
1976 Mathematics Faculty Legacy Fund Wake Forest University	2024
Starr Travel Grant Wake Forest University	2023

PUBLICATIONS

Kiendrebeogo, R. W., Farah, A. M., **Foley, E. M.**, Gray, A., Kunert, N., Puecher, A., ... & Ahumada, T. "Updated Observing Scenarios and Multimessenger Implications for the International Gravitational-wave Networks O4 and O5." *The Astrophysical Journal* 958.2 (2023): 158.
DOI: 10.3847/1538-4357/acfcb1

PRESENTATIONS

Investigating the Unruh State in 4D Schwarzschild-de Sitter Spacetime <i>Seminar presentation of honors thesis research to 50+ physics faculty members and students at Wake Forest University</i>	Winston-Salem, NC <i>Apr 2024</i>
Noise Induced Tipping in the Forced Sine Gordon Equation <i>Oral Presentation at Joint Math Meetings, 20+ attendees</i>	San Francisco, CA <i>Jan 2024</i>
Solutions to Mode Equations in 4D Schwarzschild-de Sitter Spacetime <i>Oral Presentation at American Physical Society April Meeting, 20+ attendees</i>	Minneapolis, MN <i>Apr 2023</i>
Updated Observing Scenarios Based on LIGO Public Alerts Data <i>Seminar presentation to physics department faculty and students at WFU, 50+ attendees</i>	Winston-Salem, NC <i>Aug 2022</i>
Updated Observing Scenarios Based on LIGO Public Alerts Data <i>Poster and oral presentation at REU Research symposium, 200+ attendees</i>	Minneapolis, MN <i>Aug 2022</i>
Updated Observing Scenarios Based on LIGO Public Alerts Data <i>Oral presentation given at the Zwicky Transient facility summer school workshop, 60+ attendees</i>	Minneapolis, MN <i>Jul 2022</i>

LANGUAGES AND SOFTWARE

Python, MATLAB, C++, Mathematica, HTML/CSS, Linux/Unix systems, bash, SLURM scripting, LaTeX

SERVICE AND EMPLOYMENT

Association of Women in Mathematics (AWM)

Wake Forest University

Vice President: Apr 2023 - May 2024

Secretary: Apr 2022- Apr 2023

- Published weekly newsletter to the mathematics and statistics faculty, staff, and students
- Planned weekly events to foster an inclusive and welcoming community among mathematics and statistics students at Wake Forest University.

Integrating Research in Science (IRIS) Conference Organizer

Wake Forest University

Lead Organizer: Jan 2024 - Apr 2024

Assistant Organizer: Jan 2021 - Apr 2023

- Planned a semiannual undergraduate STEM research conference with 70+ attendees in collaboration with undergraduate students at Wake Forest University and Elon University.

Mathematics Tutor

Wake Forest University

Math and Stats Center

Mar 2022 - Jan 2024

- Tutored 50+ undergraduate students for 2-4 hours weekly in the calculus sequence, linear algebra, ordinary differential equations, and complex analysis.